

MATLAB Assignment 1 [Output]

Exercise 1

```
disp("==== Exercise 1 ====");
```

```
==== Exercise 1 ====
```

```
[name, ufid, ...  
A, b, c, D, x1, x2, x3, aug, ...  
x4, x5, x6, x7, x8, ...  
F1, F2, E, m, n, E1, E2] = Exercise1()
```

```
m =  
3  
n =  
8  
E1 = 3x2  
     3     5  
     6    -1  
    -9     1  
E2 = 3x6  
     3     8     4     3     8     5  
     6     7     3     8    -1    -1  
    -9    -4     5     1     5     1  
name =  
"Evan Baesler"  
ufid =  
31151619  
A = 3x3  
     1     2     3  
     4     5     6  
    -7    -8    -9  
b = 3x1  
     2  
     4  
     6  
c = 1x3  
     2    -2     2  
D = 3x5  
     1     2     3     4     5  
    -5    -4    -3    -2    -1  
     4     3     2     1     0  
x1 = 1x3  
    -7    -8    -9  
x2 = 3x1  
     5  
    -1  
     0  
x3 = 3x4  
     1     2     3     2  
     4     5     6     4  
    -7    -8    -9     6  
aug = 4x3  
     1     2     3  
     4     5     6  
    -7    -8    -9  
     2    -2     2  
x4 = 7x7  
     1     0     0     0     0     0     0  
     0     1     0     0     0     0     0  
     0     0     1     0     0     0     0  
     0     0     0     1     0     0     0
```

```

0 0 0 0 1 0 0
0 0 0 0 0 1 0
0 0 0 0 0 0 1
x5 = 6x4
0 0 0 0
0 0 0 0
0 0 0 0
0 0 0 0
0 0 0 0
0 0 0 0
x6 = 6x6
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
x7 = 5x3
1 1 1
1 1 1
1 1 1
1 1 1
1 1 1
x8 = 3x3
2 0 0
0 -2 0
0 0 2
F1 = 3x5
8 5 3 8 4
7 -1 8 -1 3
-4 1 1 5 5
F2 = 3x5
8 4 3 8 5
7 3 8 -1 -1
-4 5 1 5 1
E = 3x8
1 2 3 8 4 3 8 5
4 5 6 7 3 8 -1 -1
-7 -8 -9 -4 5 1 5 1
m =
3
n =
8
E1 = 3x2
3 5
6 -1
-9 1
E2 = 3x6
3 8 4 3 8 5
6 7 3 8 -1 -1
-9 -4 5 1 5 1

```

```
disp("=====");
```

```
=====
```

Exercise 2

```
disp("==== Exercise 2 =====");
```

```
==== Exercise 2 =====
```

```
[~, ~, B, pivcols, compare, m, n, solution_type] = Exercise2(A, b)
```

```

aug = 3×4
    1    2    3    2
    4    5    6    4
   -7   -8   -9    6
solution_type =
"Inconsistent"
B = 3×4
    1    2    3    2
    4    5    6    4
   -7   -8   -9    6
pivcols = 1×3
    1    2    4
compare =
"rank([A]) does not equal to rank([A b])"
m =
3
n =
3
solution_type =
"Inconsistent"

```

```
disp("=====");
```

```
=====
```

Exercise 3

```
disp("==== Exercise 3 ====");
```

```
===== Exercise 3 =====
```

```

[~, ~, ...
 bA, A1, Ab1, sol1, A2, Ab2, sol2, A3, Ab3, sol3, ...
 example_A1, example_b1, example_type1, ...
 bC, A4, Ab4, sol4, A5, Ab5, sol5, A6, Ab6, sol6, ...
 example_A2, example_b2, example_type2, ...
 example_A3, example_b3, example_type3] = Exercise3()

```

```

example_aug = 2×4
    1    2    3    7
    4    5    6    8
n =
3
example_type1 =
"Consistent with Infinite Solutions"
sol4 =
"Inconsistent"
sol5 =
"Inconsistent"
sol6 =
"Inconsistent"
bA = 2×1
    4
   -2
A1 = 2×3
    4    4    6
    8    4    8
Ab1 = 2×4
    4    4    6    4
    8    4    8   -2
sol1 =
"Consistent with Infinite Solutions"
A2 = 2×3

```

```

7      5      8
4      6      4
Ab2 = 2x4
7      5      8      4
4      6      4      -2
sol2 =
"Consistent with Infinite Solutions"
A3 = 2x3
8      5      6
7      4      6
Ab3 = 2x4
8      5      6      4
7      4      6      -2
sol3 =
"Consistent with Infinite Solutions"
example_A1 = 2x3
1      2      3
4      5      6
example_b1 = 2x1
7
8
example_type1 =
"Consistent with Infinite Solutions"
bC = 3x1
-3
3
-2
A4 = 3x2
1      -1
3      4
-1      -1
Ab4 = 3x3
1      -1      -3
3      4      3
-1      -1      -2
sol4 =
"Inconsistent"
A5 = 3x2
6      0
8      3
5      -3
Ab5 = 3x3
6      0      -3
8      3      3
5      -3      -2
sol5 =
"Inconsistent"
A6 = 3x2
7      -1
7      3
6      -4
Ab6 = 3x3
7      -1      -3
7      3      3
6      -4      -2
sol6 =
"Inconsistent"
example_A2 = 3x2
1      2
3      1
4      3
example_b2 = 3x1
5
5
10
example_type2 =

```

"Consistent with One Solution"

example_A3 = 3×2

1 2

2 4

3 6

example_b3 = 3×1

3

6

9

example_type3 =

"Consistent with Infinite Solutions"

```
disp("=====");
```

=====